

Semiconductors - Asia-Pacific

Semicon Taiwan: AI drives innovation across the semiconductor stack

Estimate Change

Rising demand conviction reinforces capex & pricing in '27

We attended SEMICON Taiwan with key focus on the AI supply/demand landscape and technology innovation, with positive tone from hyperscalers, chipset design houses, foundries, OSATs and equipment/material suppliers. On firmer project pipeline in ASICs, CPUs, GPUs, networking switches and HBM base dies, we are raising our industry capex estimates for logic foundry in '27 to 23% YoY to US\$105bn and packaging and testing to 36% YoY to US\$33bn. The spending strength continues to be led by TSMC but now supplemented by commitment from OSATs. The utilization should stay high in '27 across advanced nodes (Exhibit 12) and packaging (Exhibit 15) even with aggressive expansion as chipset demand and complexity surges. Tight supply and higher equipment cost pass through should bode well for advanced technology pricing hike by 10-15%+ YoY in '27.

IC Design: accelerating chip upgrade to boost performance

Hyperscalers highlighted higher level of AI-assisted flow provided by EDA and IP vendors and their own AI systems have shortened the chipset design cycle to 9-12 months (vs. normal 1-2 years) and lowered the entry barrier of chipset customization, critical to boost system level computing power and bring down token generating cost for higher AI penetration. Google targets to launch TPU every 6-12 months and steeper production ramp in companion with their CPUs. Meta and Microsoft showcased their ASIC roadmap but would be in supplement with merchant solutions to grow computing capacity.

IC Mfg: tool modularization enables faster tech migration

On top of the trends we observed in: [our SEMICON tour](#) including CoWoS, CoPoS, SoIC, CPO/NPO, to meet customers' requirement for stronger and faster ramp of the leading edge manufacturing solutions and reduce the supply bottleneck, foundries and OSATs are collaborating for STCO (system -technology co-optimization) as the technology development is shifting from serial to parallel across chip production, package, materials and system players. Specifically, TSMC noted the importance of higher level of back-end tool automation and modularization as it could reduce time for re-validation and provide the supply chain with higher level of fungibility for different generations of technologies.

Preference for higher AI manufacturing and ASIC leverage

Consistent with OSAT and tool makers' feedback, TSMC's comment at SEMICON that stronger demand has led to a rise in the equipment orders it needs to place which has grown from 1x by the end of 2025 to 1.5x in 1Q26 and further to 1.9x by July. As AI capex continues on technology upgrade, we keep preference on the companies with higher exposure in CPU/ASIC/networking switches and CPO optionality and lifting PO and estimates on TSMC ASE, All Ring, Mediatek while stay positive on Chroma and GUC.

08 September 2026

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See page 18 for glossary

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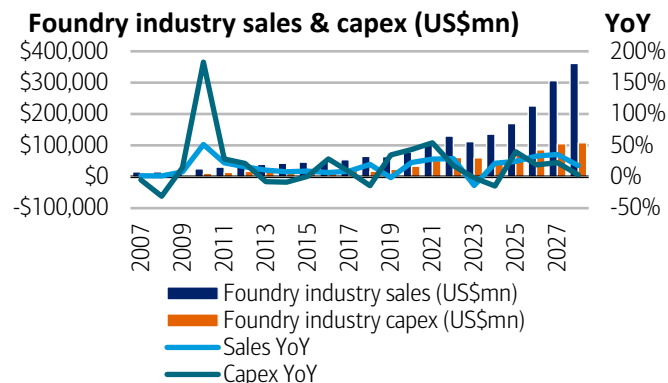
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Focus charts

Exhibit 1: Front-end sales and capex YoY

Global front-end industry Capex will grow at 12% CAGR over 2026-28

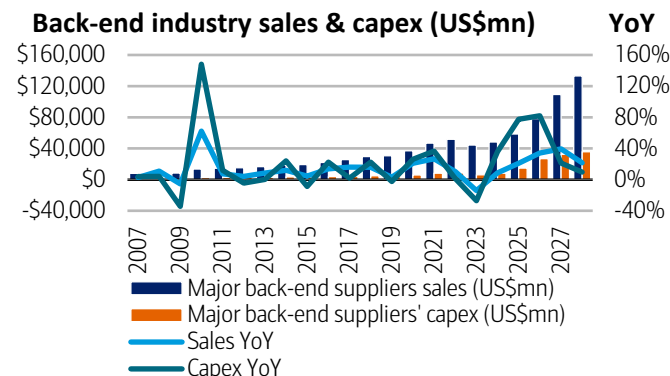


Source: Company Data, BofA Global Research estimates, Bloomberg

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Exhibit 2: Back-end sales and capex YoY

Global back-end industry Capex will grow at 15% CAGR over 2026-28

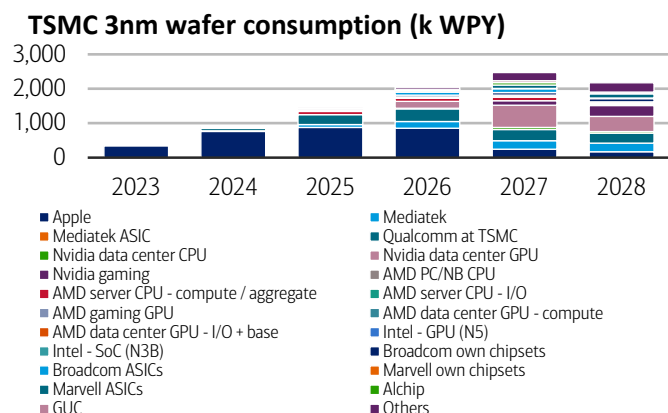


Source: Company Data, BofA Global Research estimates, Bloomberg

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Exhibit 3: TSMC 3nm wafer consumption

Nvidia will become the largest contributor from 2027

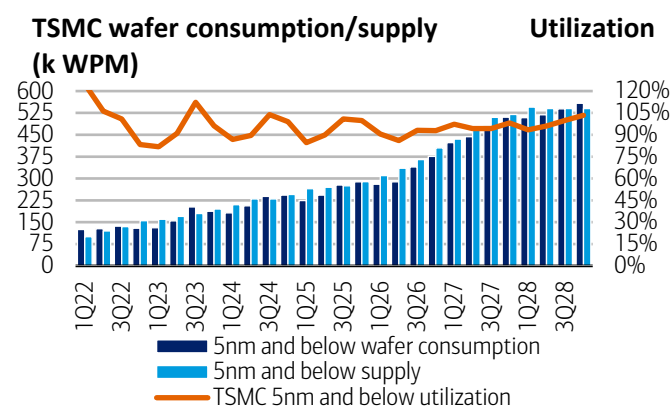


Source: BofA Global Research estimates

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Exhibit 4: TSMC 5nm and below supply/demand and utilization

5nm and below utilization remains close to full through 2028

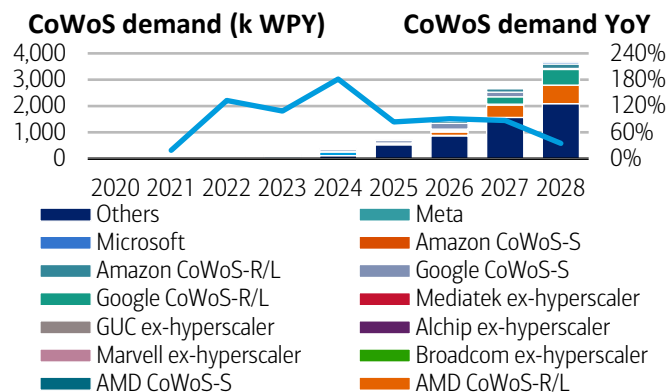


Source: BofA Global Research estimates

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Exhibit 5: CoWoS demand breakdown by customers

CoWoS demand should remain strong through 2028

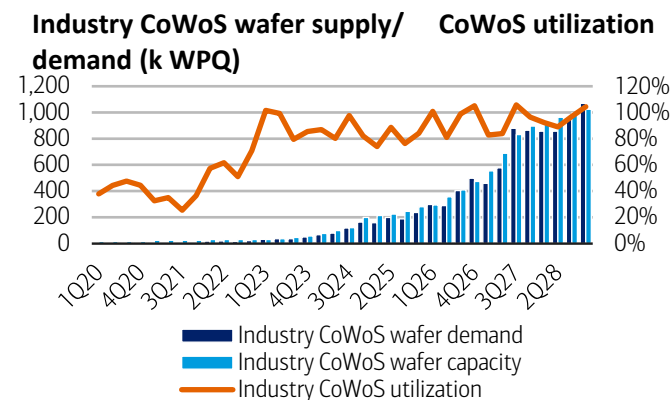


Source: BofA Global Research estimates

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Exhibit 6: Industry CoWoS supply/demand

CoWoS utilization should stay above 90% through 2028



Source: BofA Global Research estimates

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Exhibit 7: Taiwan semiconductor coverage valuation table

Our coverage valuation table

		Share		Mkt cap		EPS (LC)		PER (X)		PBR (X)		Div. yield (%)		ROE (%)		EV/EBITDA	
Ticker	Company	price (LC)	Rating	(US\$m)	2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E	
Foundry																	
2330 TT	TSMC	2,460.0	BUY	2,022,049	110.20	155.85	22.3	15.8	8.1	5.6	1.0	1.0	43.0	42.0	14.8	10.6	
2303 TT	UMC	143.0	U/P	56,976	7.37	6.05	19.4	23.6	4.0	3.7	2.0	2.0	22.3	16.2	14.4	11.5	
5347 TT	Vanguard	156.0	BUY	9,289	6.43	10.68	24.3	14.6	5.6	4.3	2.9	2.9	20.7	33.3	17.6	10.5	
3105 TT	Win Semi	450.5	U/P	6,054	8.59	11.11	52.4	40.6	4.2	4.0	0.0	0.0	8.4	10.2	30.9	26.6	
6770 TT	PSMC	72.9	BUY	10,888	6.78	7.50	10.8	9.7	2.5	2.0	0.0	0.0	28.5	23.0	7.8	6.5	
Foundry average								25.8	20.9	4.9	3.9	1.2	1.2	24.6	24.9	17.1	13.1
OSAT																	
3711 TT	ASE	621.0	BUY	87,997	18.13	30.90	34.3	20.1	5.8	4.3	1.8	3.1	20.3	25.1	17.4	10.2	
2449 TT	KYEC	273.0	BUY	11,110	8.38	16.59	32.6	16.5	4.6	3.4	3.3	1.7	16.9	23.6	15.3	9.0	
OSAT average								33.4	18.3	5.2	3.8	2.6	2.4	18.6	24.4	16.4	9.6
Fabless																	
5269 TT	ASMedia	1,440.0	Neutral	3,385	95.39	108.64	15.1	13.3	3.1	3.7	3.8	4.5	20.2	25.5	32.8	28.9	
2454 TT	MediaTek	4,760.0	BUY	242,149	68.61	153.19	69.4	31.1	17.5	12.0	1.2	2.6	26.1	45.5	56.3	24.6	
3034 TT	Novatek	540.0	Neutral	10,415	34.63	37.03	15.6	14.6	4.4	4.1	4.3	5.0	29.6	29.1	11.8	10.7	
6415 TT	Silergy	418.0	BUY	5,160	13.55	20.32	30.9	20.6	3.9	3.4	1.3	1.9	13.3	17.7	31.0	17.0	
5274 TT	Aspeed	19,215.0	BUY	25,326	203.84	345.36	94.3	55.6	58.7	35.4	0.7	1.0	78.7	79.4	76.8	43.8	
4966 TT	Parade	566.0	U/P	1,417	31.83	36.23	17.8	15.6	2.0	1.9	2.8	3.2	11.3	12.3	10.6	9.8	
2379 TT	Realtek	729.0	Neutral	11,912	27.20	32.48	26.8	22.4	9.7	8.7	3.2	3.8	37.0	40.9	17.9	15.3	
3035 TT	Faraday	179.0	U/P	1,478	4.05	7.07	44.2	25.3	2.4	2.2	1.6	2.8	6.6	9.2	18.1	12.5	
3443 TT	GUC	5,935.0	BUY	25,210	47.62	91.68	124.6	64.7	47.8	34.9	0.5	0.8	43.2	62.3	99.0	51.9	
3529 TT	eMemory	2,570.0	BUY	6,084	35.40	49.12	72.6	52.3	36.8	29.3	1.2	1.6	58.7	62.4	59.3	44.0	
3661 TT	Alchip	4,040.0	BUY	11,052	132.94	204.17	30.4	19.8	4.9	4.2	1.7	2.6	20.2	23.0	21.4	14.4	
6643 TT	M31	411.0	U/P	555	1.15	6.73	358.3	61.1	8.6	7.3	0.2	1.2	2.5	12.9	165.7	38.7	
Fabless average								70.9	32.3	15.9	11.8	1.8	2.5	29.7	34.4	47.7	25.3
Semiconductor equipment / consumables suppliers																	
6488 TT	GlobalWafers	971.0	U/P	14,715	21.51	29.58	45.1	32.8	4.6	4.2	1.0	1.9	10.6	13.4	31.3	21.4	
3131 TT	Grand Process Technology Corp	2,555.0	BUY	2,370	70.37	117.05	36.3	21.8	14.8	10.7	2.0	3.3	41.5	56.8	33.1	17.4	
2360 TT	Chroma	2,230.0	BUY	30,058	46.73	80.05	47.7	27.9	26.6	17.0	0.9	1.5	45.1	45.7	41.1	23.2	
7769 TT	Hon Precision	6,565.0	BUY	37,441	123.54	178.95	53.1	36.7	17.3	14.5	1.5	2.2	35.2	43.0	41.8	28.4	
6187 TT	All Ring	1,370.0	BUY	4,264	28.94	49.18	n.a.	n.a.	n.a.	n.a.	1.3	2.2	33.0	40.9	43.9	24.7	
1560 TT	Kinik	730.0	BUY	3,477	15.71	25.38	46.5	28.8	11.5	9.0	0.7	1.1	26.5	35.0	34.2	22.0	
6515 TT	Winway	7,000.0	BUY	8,086	105.78	210.72	66.2	33.2	32.6	19.3	0.9	1.7	53.5	73.1	51.6	24.7	
6223 TT	MPI	5,490.0	BUY	17,050	63.61	125.84	86.3	43.6	31.1	19.9	0.7	1.4	39.1	55.6	65.8	34.3	
Semiconductor equipment / consumables average								54.5	32.1	19.8	13.5	1.1	1.9	35.6	45.4	42.8	24.5
Semis distributor																	
3036 TT	WT Micro	192.5	BUY	7,752	26.62	31.51	7.2	6.1	2.0	1.8	9.7	11.5	29.3	30.7	5.0	3.9	
Semis distributor average								7.2	6.1	2.0	1.8	9.7	11.5	29.3	30.7	5.0	3.9

Source: BofA Global Research estimates, Bloomberg

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SEMICON Taiwan: AI pushes technology frontier and supply chain to its limits

We attended SEMICON Taiwan held from August 31st to September 4th, featuring all time high 1,300+ exhibitors, 4,300+ booths and 100k+ professional visitors, suggesting an even higher interest level across the supply chain joining the event to showcase their technology roadmap and product solutions with their customers and prospective partners amid strong semiconductor industry growth outlook across advanced logic and memory amid booming cloud AI demand.

Importantly, senior executives and technical experts from the manufacturing supply chain (TSMC, GlobalFoundries, UMC, ASE, Powertech KYEC, Globalwafers, Unimicron, Applied Materials, Tokyo Electron, etc.), hyperscalers (Google, Meta, Microsoft) and IC design companies (e.g. Nvidia, AMD, Broadcom, Marvell, Mediatek, Synopsys, Cadence, Arya Labs) presented their latest views on advanced logic, 2.5D/3D packaging, testing and metrology, silicon photonics and AI-enabled chipset design.

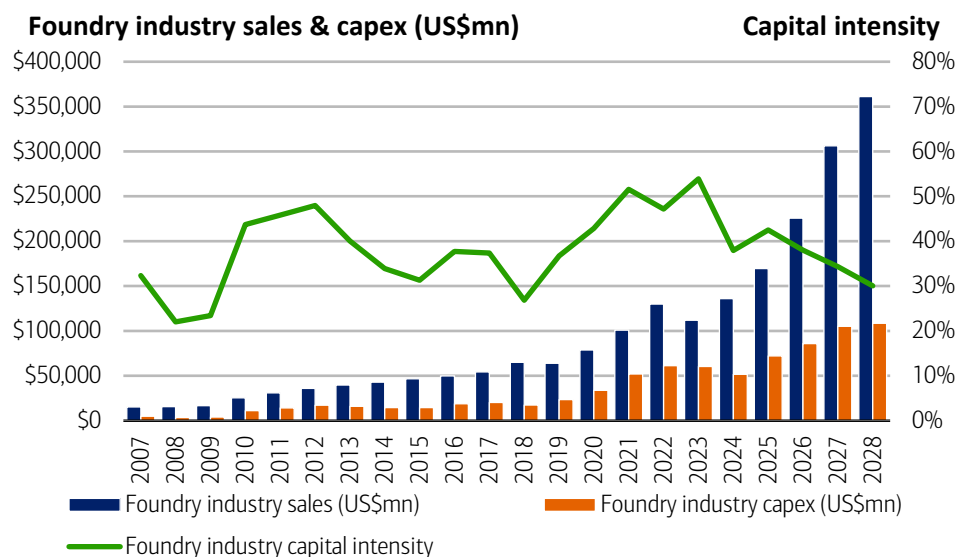
We profile our takeaways from SEMICON Taiwan with valuable feedback across the supply chain on how accelerating AI product cycles are reshaping capacity requirements, technology roadmaps and manufacturing bottlenecks. Specifically, we highlight the industry updates we learnt from the expo and keynotes which reinforce our view of stronger for longer AI chipset demand outlook requiring even more aggressive capacity expansion and technology upgrades in 2027-28 across the front-end and back-end manufacturing supply chain to catch up with the customers' requirements.

In this report, we are updating our industry models including

1. **The logic foundry industry capex outlook.** We lift the growth estimates from 9%/26% YoY to 19%/23% YoY for '26/27 to US\$86bn/US\$105bn, supporting a CAGR of 12% from '26-28 (vs. 16% prior). The upward revision is mostly led by our higher expectation on TSMC's capex ramping from US\$62bn in '26 to US\$80bn in '27 on stronger wafer demand on N3 and N2 from GPUs, ASICs, CPUs, networking switches and HBM base dies and initial ramp of A16. We already lifted UMC and Vanguard's capex expectations through 2Q26 earnings season based on the companies' latest guidance.

Exhibit 8: Front-end sales, capex, and capital intensity

Front-end capex should reach US\$86bn/US\$105bn in '26/27



Source: Company Data, BofA Global Research estimates, Bloomberg

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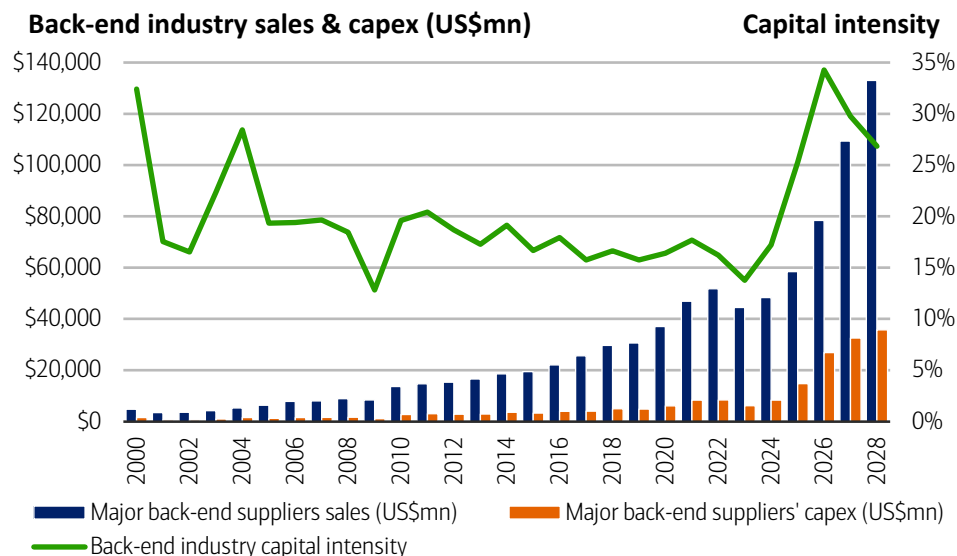


- 2. The back-end industry capex outlook.** We raise the back-end packaging and testing industry spending estimates from 66%/36% YoY to 82%/21% YoY for '26/27 to US\$27bn/US\$33bn, supporting a CAGR of 15% from '26-28 (vs. 23% prior). The upward revision is driven by our higher expectation of TSMC's more aggressive capacity addition on CoWoS-R/L to accommodate more demand from its customers amid growing level of chipset complexity and reticle size and higher level of capacity readiness of SoIC for higher 3D packaging adoption for logic and memory dies integration and CPO.

In addition, the OSATs are accelerating their investment for the 2.5D packaging (e.g. ASE's FoCoS and panel level packaging, Amkor's SWIFT and Powertech's panel level solutions) led by aggressive expansion by ASE across advanced wafer and chipset packaging and testing.

Exhibit 9: Back-end sales, capex, and capital intensity

Front-end capex should reach US\$27bn/US\$33bn in '26/27



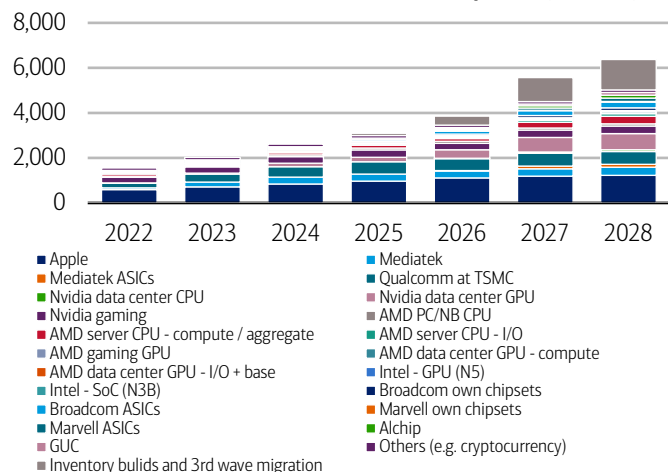
Source: Company Data, BofA Global Research estimates, Bloomberg

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- 3. Advanced node supply/demand across 3nm, 2nm and A14.** We are raising our expectations mildly. With 3nm capacity ramping from 190k WPM by 4Q26 to 235k WPM as we believe TSMC tries to convert more 5nm capacity to support solid 3nm demand in addition to the ongoing greenfield and brownfield expansion in Taiwan and overseas. With stronger than expected pipeline from its CPU and ASIC customers beyond the high-end mobile applications, we expect TSMC to also pull the ramp schedule of its 2nm capacity from 90k WPM in 4Q26 to 200k by 4Q27 while gets ready for A14 node initial installed base.

Exhibit 10: TSMC 5nm and below wafer consumption breakdown

5nm and below demand should remain strong, supported by CPU and ASIC customers beyond the high-end mobile applications

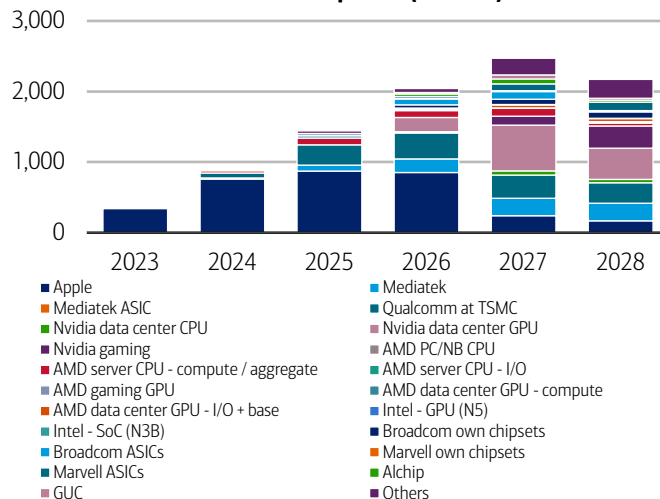
TSMC 5nm and below wafer consumption (k WPY)

Source: BofA Global Research estimates

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Exhibit 11: TSMC 3nm wafer consumption breakdown

Nvidia should become the largest contributor in TSMC 3nm from 2027

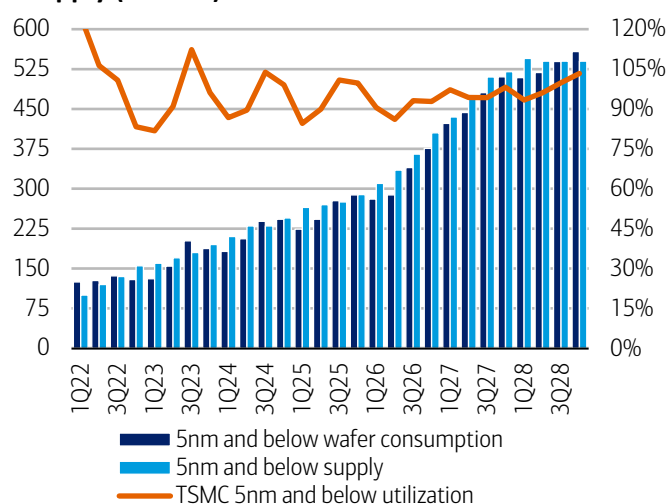
TSMC 3nm wafer consumption (k WPY)

Source: BofA Global Research estimates

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Exhibit 12: TSMC 5nm and below supply/demand and utilization

5nm and below utilization should remain close to full through 2028

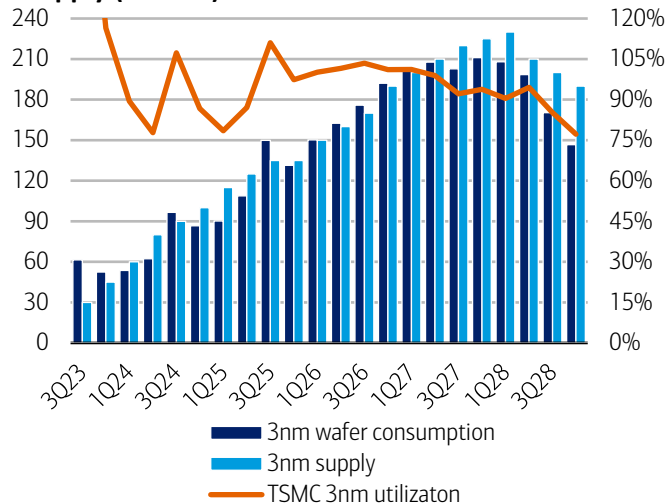
TSMC wafer consumption/ supply (k WPM)

Source: BofA Global Research estimates

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Exhibit 13: TSMC 3nm supply/demand and utilization

3nm utilization should remain above 90% through 2027

TSMC wafer consumption/ supply (k WPM)

Source: BofA Global Research estimates

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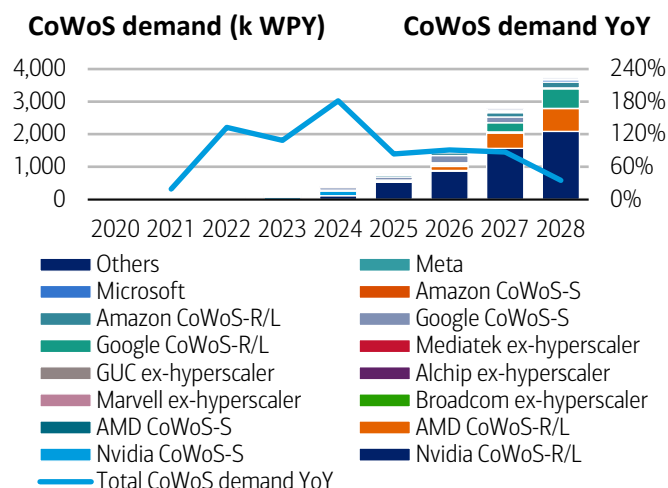
- 4. Supply/demand model for advanced 2.5D packaging (CoWoS, FoCoS, etc.).** Similar to our recent supply chain feedback, we are lifting the industry CoWoS (and similar) supply growth along with our industry capex model to 280k WPM by 4Q27 (vs. 270k prior), growing from 160k in 4Q26 (unchanged).

In addition, based on the industry discussions at SEMICON Taiwan, we expect the commercialization of panel level packaging is skewed toward late-2028 or early-2029. Instead of becoming mainstream solution, the technology would be supplemental in the high-end applications to the existing CoWoS solutions which is extending its reticle size support to more than 14x in 2029.



Exhibit 14: CoWoS demand breakdown by customers

CoWoS demand should grow at 59% CAGR from 2026-28

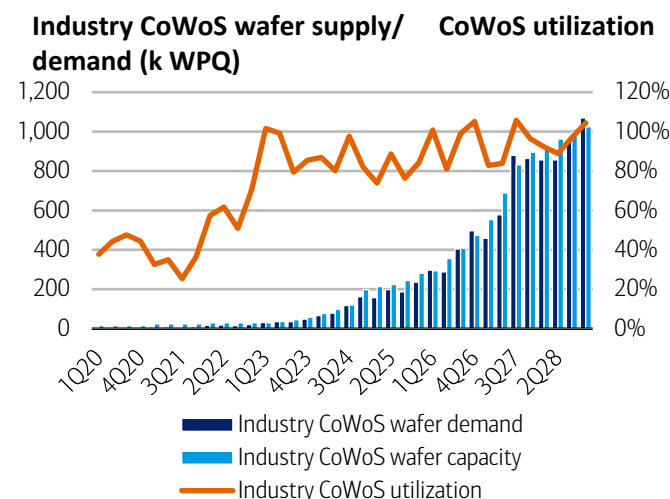


Source: BofA Global Research estimates

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Exhibit 15: Industry CoWoS supply/demand

CoWoS utilization should stay above 90% through 2028



Source: BofA Global Research estimates

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We are lifting our earnings expectations for TSMC and ASE to reflect a stronger than feared AI demand outlook and potentially better Apple product cycle even factoring in higher depreciation from more aggressive capex spending.

We raise our 2027/28 EPS estimates for TSMC from NT\$153.17/181.96 to NT\$155.85/187.90 as we factor in stronger AI chipsets demand across advanced front-end and back-end. We maintain our Buy rating with PO at NT\$3,100 on 20x '27 P/E. The share valuation is undemanding at 16x '27 P/E on strong technology leadership and extending AI visibility.

Exhibit 16: TSMC 2026-28 estimates BofAe vs. street

We lift our 2027/28 EPS to NT\$155.85/187.90

	2026E				2027E				2028E			
	BofAe	Old	Street	vs. street	BofAe	Old	Street	vs. street	BofAe	Old	Street	vs. street
Sales (NT\$mn)	\$5,477,153	\$5,477,153	\$5,428,015	1%	\$7,711,575	\$7,624,785	\$7,286,032	6%	\$9,150,819	\$8,920,142	\$9,118,434	0%
Sales QoQ / YoY (%)	43.8%	43.8%	42.5%		40.8%	39.2%	34.2%		18.7%	17.0%	25.1%	
Gross profit (NT\$mn)	\$3,654,584	\$3,654,584	\$3,617,935	1%	\$5,191,137	\$5,107,101	\$4,830,348	7%	\$6,172,734	\$5,989,971	\$6,018,348	3%
GMs (%)	66.7%	66.7%	66.7%	0%	67.3%	67.0%	66.3%		67.5%	67.2%	66.0%	
Operating profit (NT\$mn)	\$3,251,539	\$3,251,539	\$3,185,005	2%	\$4,689,032	\$4,604,996	\$4,252,251	10%	\$5,606,193	\$5,423,431	\$5,269,144	6%
OpMs (%)	59.4%	59.4%	58.7%	1%	60.8%	60.4%	58.4%	2%	61.3%	60.8%	57.8%	3%
Pretax profit (NT\$mn)	\$3,446,566	\$3,446,566	\$3,366,090	2%	\$4,870,519	\$4,786,823	\$4,413,051	10%	\$5,872,206	\$5,686,699	\$5,494,673	7%
Pretax margins (%)	62.9%	62.9%	62.0%	1%	63.2%	62.8%	60.6%	3%	64.2%	63.8%	60.3%	4%
Tax expense (NT\$mn)	-\$588,648	-\$588,648	-\$582,044	1%	-\$828,896	-\$814,665	-\$753,534	10%	-\$999,545	-\$967,941	-\$922,475	8%
Tax rate (%)	17.1%	17.1%	17.3%	0%	17.0%	17.0%	17.1%	0%	17.0%	17.0%	16.8%	0%
Net profit (NT\$mn)	\$2,857,919	\$2,857,919	\$2,784,046	3%	\$4,041,623	\$3,972,158	\$3,659,517	10%	\$4,872,661	\$4,718,757	\$4,572,198	7%
Net margins (%)	52.2%	52.2%	51.3%	1%	52.4%	52.1%	50.2%	2%	53.2%	52.9%	50.1%	3%
EPS (NT\$)	\$110.21	\$110.21	\$106.89	3%	\$155.85	\$153.17	\$139.25	12%	\$187.90	\$181.96	\$174.30	8%

Source: BofA Global Research estimates, Bloomberg

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We raise our 2027/28 EPS estimates for ASE from NT\$29.80/42.59 to NT\$30.90/45.30 as we factor in stronger advanced packaging demand and improving mainstream business. We rate ASE Buy and raise PO from NT\$750 to NT\$800 on unchanged 20x 2H27-1H28 P/E. While ASE may need fund raising for its capex in '27-28, it could secure the customer commitment while ROE from LEAP business at 25-30%+ should justify the cost of capital from the fundraising.

Exhibit 17: ASE 2026-28 estimates BofAe vs. street

We lift our 2027/28 EPS to NT\$30.90/45.30

	2026E				2027E				2028E			
	BofAe	Old	Street	vs. street	BofAe	Old	Street	vs. street	BofAe	Old	Street	vs. street
Sales (NT\$mnn)	\$846,543	\$846,543	\$789,103	7%	\$1,089,719	\$1,077,953	\$956,321	14%	\$1,297,628	\$1,270,507	\$1,130,478	15%
Sales QoQ / YoY (%)	31.2%	31.2%	22.3%		28.7%	27.3%	21.2%		19.1%	17.9%	18.2%	
Gross profit (NT\$mnn)	\$179,917	\$179,917	\$166,540	8%	\$290,641	\$283,689	\$222,153	31%	\$396,978	\$380,257	\$278,968	42%
GMs (%)	21.3%	21.3%	21.1%	0%	26.7%	26.3%	23.2%	3%	30.6%	29.9%	24.7%	6%
Operating profit (NT\$mnn)	\$98,585	\$98,585	\$93,225	6%	\$192,658	\$186,311	\$138,830	39%	\$286,762	\$271,374	\$186,338	54%
OpMs (%)	11.6%	11.6%	11.8%	0%	17.7%	17.3%	14.5%	3%	22.1%	21.4%	16.5%	6%
Pretax profit (NT\$mnn)	\$101,997	\$101,997	\$94,219	8%	\$175,866	\$169,519	\$139,333	26%	\$257,460	\$242,072	\$180,488	43%
Pretax margins (%)	12.0%	12.0%	11.9%	0%	16.1%	15.7%	14.6%	2%	19.8%	19.1%	16.0%	4%
Tax expense (NT\$mnn)	-\$20,949	-\$20,949	-\$19,838	6%	-\$36,619	-\$35,262	-\$28,600	28%	-\$53,357	-\$50,165	-\$35,928	49%
Tax rate (%)	20.5%	20.5%	21.1%	-1%	20.8%	20.8%	20.5%	0%	20.7%	20.7%	19.9%	1%
Net profit (NT\$mnn)	\$81,048	\$81,048	\$74,380	9%	\$139,247	\$134,257	\$110,732	26%	\$204,103	\$191,907	\$144,561	41%
Net margins (%)	9.6%	9.6%	9.4%	0%	12.8%	12.5%	11.6%	1%	15.7%	15.1%	12.8%	3%
EPS (NT\$)	\$18.13	\$18.13	\$16.83	8%	\$30.90	\$29.80	\$25.00	24%	\$45.30	\$42.59	\$32.90	38%

Source: BofA Global Research estimates, Bloomberg

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In the equipment supply chain, we are also raising our 2027/28 EPS estimates for All Ring from NT\$44.20/54.56 to NT\$49.18/54.90 amid solidifying business pipeline into 2027 with firmer order visibility well backed by the accelerated 2.5D packaging capacity expansion across the industry and CPO optionality driving its sales growth and operating leverage.

We keep Buy rating and raise PO from NT\$1,550 to NT\$1750 based on unchanged 36x 2027E P/E (vs. its 8-41x range) on stronger 2.5D packaging expansion and CPO optionality.

Exhibit 18: All Ring 2026-28 estimates BofAe vs. street

We lift our 2027/28 EPS to NT\$49.18/54.90

	2026E				2027E				2028E			
	BofAe	Old	Street	vs. street	BofAe	Old	Street	vs. street	BofAe	Old	Street	vs. street
Sales (NT\$mnn)	\$9,359	\$9,359	\$9,241	1%	\$15,596	\$14,074	\$14,080	11%	\$17,198	\$16,882	\$19,262	-11%
Sales QoQ / YoY (%)	74.4%	74.4%	72.2%		66.6%	50.4%	52.4%		10.3%	20.0%	36.8%	
Gross profit (NT\$mnn)	\$5,009	\$5,009	\$4,842	3%	\$8,401	\$7,568	\$7,507	12%	\$9,321	\$9,147	\$10,401	-10%
GMs (%)	53.5%	53.5%	52.4%	1%	53.9%	53.8%	53.3%	1%	54.2%	54.2%	54.0%	0%
Operating profit (NT\$mnn)	\$2,834	\$2,834	\$2,943	-4%	\$5,097	\$4,526	\$4,969	3%	\$5,705	\$5,673	\$6,794	-16%
OpMs (%)	30.3%	30.3%	31.8%	-2%	32.7%	32.2%	35.3%	-3%	33.2%	33.6%	35.3%	-2%
Pretax profit (NT\$mnn)	\$3,290	\$3,290	\$3,097	6%	\$5,601	\$5,031	\$5,098	10%	\$6,253	\$6,214	\$7,235	-14%
Pretax margins (%)	35.2%	35.2%	33.5%	2%	35.9%	35.7%	36.2%	0%	36.4%	36.8%	37.6%	-1%
Tax expense (NT\$mnn)	-\$476	-\$476	-\$539	-12%	-\$818	-\$733	-\$905	-10%	-\$914	-\$908	-\$1,366	-33%
Tax rate (%)	14.5%	14.5%	17.4%	-3%	14.6%	14.6%	17.8%	-3%	14.6%	14.6%	18.9%	-4%
Net profit (NT\$mnn)	\$2,814	\$2,814	\$2,558	10%	\$4,783	\$4,298	\$4,193	14%	\$5,339	\$5,306	\$5,868	-9%
Net margins (%)	30.1%	30.1%	27.7%	2%	30.7%	30.5%	29.8%	1%	31.0%	31.4%	30.5%	1%
EPS (NT\$)	\$28.94	\$28.94	\$28.28	2%	\$49.18	\$44.20	\$43.35	13%	\$54.90	\$54.56	\$60.64	-9%

Source: BofA Global Research estimates, Bloomberg

BofA GLOBAL RESEARCH

For fabless, we are raising our 2027 EPS for Mediatek from NT\$137.76 to NT\$153.19 mainly led by its cloud ASIC business with its stronger confidence in Google's TPU business ramping from 4Q26 through 2028 for the next few generation projects, more than offsetting still muted Android smartphone market demand in the next few quarters.

With the investment from Nvidia/Alphabet reinforcing our view on Mediatek's strategic ASIC opportunity, we keep our Buy rating and PO at NT\$5,230 on 23x 2H27-1H28E P/E. We believe the shares are attractive at 14x '28E P/E, backed by its solid sales/earnings outlook in '26-28E at 84%/125% CAGR as operating leverage could accelerate backed by cloud/auto ASIC projects ramp and potential in edge AI with strong smart edge solutions.



Exhibit 19: Mediatek 2026-28 estimates BofAe vs. street

We now model 2027/28 EPS to be NT\$153.19/347.12

	2026E				2027E				2028E			
	BofAe	Old	Street	vs. street	BofAe	Old	Street	vs. street	BofAe	Old	Street	vs. street
Sales (NT\$mnn)	\$667,794	\$667,794	\$651,622	2%	\$1,238,800	\$1,124,645	\$1,061,185	17%	\$2,255,967	\$2,255,967	\$1,875,319	20%
Sales QoQ / YoY (%)	12.1%	12.1%	9.3%		85.5%	68.4%	62.9%		82.1%	100.6%	76.7%	
Gross profit (NT\$mnn)	\$303,921	\$303,921	\$298,345	2%	\$539,892	\$494,230	\$471,134	15%	\$951,942	\$951,942	\$815,164	17%
GMs (%)	45.5%	45.5%	45.8%	0%	43.6%	43.9%	44.4%	-1%	42.2%	42.2%	43.5%	-1%
Operating profit (NT\$mnn)	\$105,347	\$105,347	\$104,214	1%	\$269,723	\$240,797	\$223,045	21%	\$624,881	\$625,626	\$466,893	34%
OpMs (%)	15.8%	15.8%	16.0%	0%	21.8%	21.4%	21.0%	1%	27.7%	27.7%	24.9%	3%
Pretax profit (NT\$mnn)	\$127,080	\$127,080	\$121,682	4%	\$285,000	\$256,094	\$242,267	18%	\$645,971	\$646,094	\$481,628	34%
Pretax margins (%)	19.0%	19.0%	18.7%	0%	23.0%	22.8%	22.8%	0%	28.6%	28.6%	25.7%	3%
Tax expense (NT\$mnn)	-\$17,872	-\$17,872	-\$16,582	8%	-\$41,167	-\$36,815	-\$36,921	12%	-\$93,455	-\$93,370	-\$73,940	26%
Tax rate (%)	14.1%	14.1%	13.6%	0%	14.4%	14.4%	15.2%	-1%	14.5%	14.5%	15.4%	-1%
Net profit (NT\$mnn)	\$109,208	\$109,208	\$105,101	4%	\$243,832	\$219,279	\$205,346	19%	\$552,515	\$552,724	\$407,688	36%
Net margins (%)	16.4%	16.4%	16.1%	0%	19.7%	19.5%	19.4%	0%	24.5%	24.5%	21.7%	3%
EPS (NT\$)	\$68.61	\$68.61	\$66.29	4%	\$153.19	\$137.76	\$130.35	18%	\$347.12	\$347.25	\$254.48	36%

Source: BofA Global Research estimates, Bloomberg

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Takeaways on the major industry discussion

IC design:

Hyperscalers spoke at SEMICON Taiwan – Google, Microsoft and Meta – have spent the past year pushing custom silicon programs specifically to compress design cycles and reduce merchant AI accelerator chipset dependency.

Google positioned cloud companies as active architects of custom silicon, data center systems and networks and noted AI infrastructure has shifted from compute-limited to memory-limited, with HBM and advanced DRAM now representing 75% of AI server BOM cost, requiring the hyperscalers to accelerate the chipset development and expand ASIC pipeline to maximize the resource they have.

Exhibit 20: Google's "square wave" framework

"Square wave" framework now commits massive ASIC production at tape out before months of validation



Source: Google

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Instead of continuing the old model of running the initial silicon testing for its generation TPUs before committing to volume production which takes 3-6 months, Google through its "square wave" framework now commits to massive ASIC production

at tape out before months of validation, allowing the chipset deployment at full scale as soon as it can while reducing the risk of supply situation changes in the chain where supply shortage becomes a common issue. To enable this, Google is targeting to collaborate with its partners spanning from chipset test/validation, software and performance verification and AI system co-design and manufacturing.

Exhibit 21: Synopsys's AI assisted design flow

Synopsys's AI assisted design flow accelerates chipsets design



Source: Synopsys

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In contrast to Google's aggressive push on deploying its own silicon, Meta noted its hardware portfolio approach across its own MTIA400, Nvidia's GB300 and AMD's MI455x deployed together rather than fully replacing merchant solutions as it believes the workload is getting more diversified across training, inference prefill, inference decode, recommender training and recommender inference.

The hyperscalers' comments on shortening ASIC cadence to accelerate AI computing capability upgrades has been evidenced by the recent development in OpenAI's Jalapeno, unveiled with Broadcom as a custom inference ASIC. The chipset from the initial design to manufacturing tape out in only 9 months, one of the fastest ASIC development cycles achieved at this level of sophistication and is targeted to start small production by the end of 2026 and scale through 2027 as part of a 10GW multi generation infrastructure commitment for OpenAI's own use.

Notably, on the system level, the hyperscalers addressed the reliability issue especially on networking, which they see 70% of the AI cluster hardware failures is related to scale up/out networking (vs. 15%/15% from host/chipset issues). Therefore, the stability and scalability of networking solutions would be critical in 2027-28 to enable more efficient AI computing model upgrades.

Front-end: share gains in grinding process & SiC wafer

Manufacturing: TSMC is accelerating its capacity expansion fivefold this year but still is not able to keep up with AI demand. According to the company's comment in the fireside chat on CEO Summit at SEMICON Taiwan, in addition to deploying AI heavily in fabs with productivity and yield gains, it is building 25 front-end manufacturing fabs and advanced packaging facilities worldwide (vs. 5-6 per annum historically), with 13 of them in Taiwan. However, the ecosystem was not ready, including the construction labor shortage while the equipment delivery schedule is also extended meaningfully due to under supply.

Specifically, to share its view on the surging AI demand, TSMC provides a reference that if its equipment orders for 2026 to meet its customers demand was 1x by the end of 2025, the orders rose to 1.5x after 1 quarter and further reached 1.9x of the original

expectation by July this year, suggesting its equipment demand grew by 50% in 1Q26 vs. 4Q25, and was lifted by another ~23% by July. The prolonged equipment delivery lead time along with the faster technology migration have been prompting TSMC to request its materials and equipment supply chain introducing more modular, cross-generation common platforms

Materials: Kinik showcased its solutions across its pad conditioner with 70%+/80% share in TSMC's N3/N2 process with growing content as CMP step count rises along with the node migration especially for the backside power delivery and 2.5D/3D packaging and hybrid bonding. Its carrier wafer demand for Micron's HBM is also tracking strongly. Importantly, it demonstrated its grinding wheels with solid qualification progress at its major foundry customer and could start contribution from 2027 at the expense of its overseas competitors.

Separately, with growing heat dissipation requirement for HPC/AI chipsets, SiC carrier wafer (not interposer) will likely be in '28 for new generation GPU. Such business could be 15-20% of GWC's revenue based on the assumption of 1) 10-20kwpm capacity; and 2) US\$2k+ ASP. The company is preparing corresponding capacity in Zhunan, with mass shipment kicking off from late-2027 at earliest. The company has also demonstrated its 310x310mm square wafer solutions for panel level packaging.

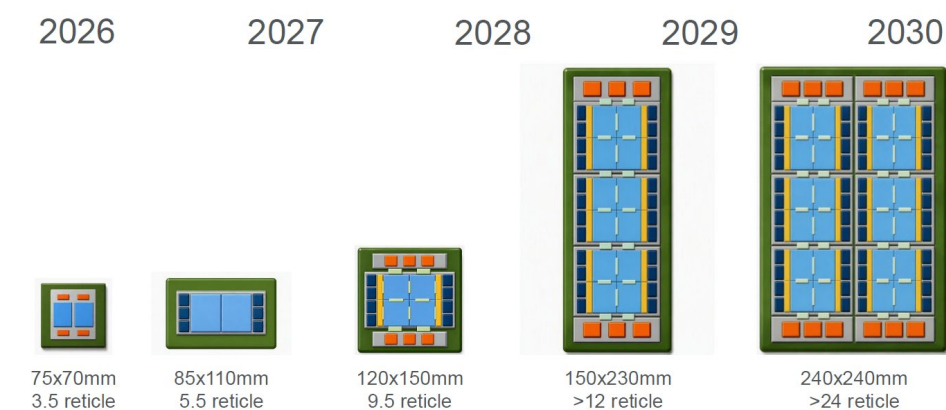
2.5D packaging: solid technology and supply expansion

TSMC: the company largely maintains its technology roadmap on CoWoS it shared earlier this year at its technology Symposiums (see our update in: [May](#) and: [June](#)) that it will have CoWoS solutions to support more than 14x reticle size package by '29 to support much faster I/O on higher bandwidth and package power up to 4,100W while reduce power delivery loss by 5x. On the SoIC, at the SEMICON Taiwan keynotes, it expects to see 11 tape outs for GPU/CPU by 4Q26 which we believe will require much more capacity support than the current 15-20k WPM capacity in the next few years.

ASE: the company demonstrated its 2.5D packaging roadmap on FoCoS which can support up to 5.5x reticle size package with HBM4 integrated and targets to extend the technology offering to 7.5x/9.5x reticle size applications by 2027/30 respectively. If the migration and yield improvement are well executed, we believe this should allow the company to win the business from its customers directly in inference ASICs, CPUs and networking switches in addition to the outsourcing orders from TSMC.

Exhibit 22: ASE's panel level packaging roadmaps

ASE plans the mini production on 300x310mm panel technology in 2027 with higher equipment commonality



Source: ASE

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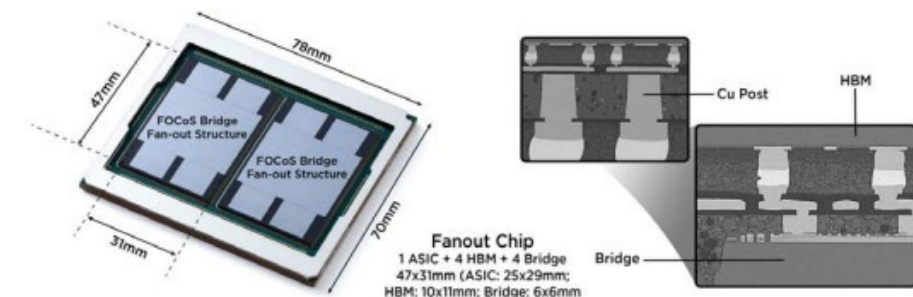
At the panel level packaging forum, ASE noted the 300x310mm panel productivity is 1.78x higher than wafer level packaging for the packages with 5.5x+ reticle size and could be up to 2.25x for 9.5x reticle size packages. As the benefit scales along with the

size of packages, the company believes the technology will see more traction and it already demoed 3.0x reticle size panel level packaging technology with good results.

While it plans the mini production on 300x310mm panel technology in 2027 with higher equipment commonality vs. wafer level solutions, ASE does not expect the solution to contribute meaningfully until 2029 and sees the larger size panel upgrade takes even more time due to lack of industry standard on the process (e.g. spin coating vs. split coating) and lack of mature ecosystem of the material and equipment supply.

Exhibit 23: ASE's FoCoS-Bridge technology roadmap

Gen 2 of FoCoS-Bridge could support 2x ASIC with 8x HBMs



Body Size

FOCoS-Bridge Device	1 st Generation	2 nd Generation
Configuration	1 x ASIC + 1 x HBM	2 x ASIC + 8 x HBM
Fan-Out Size (mm)	27 x 15	(47 x 31) x 2
Package Size (mm)	41 x 31	78 x 70
RDL L/S (um)/ RDL Layer Count	10/10, 1 Layer	10/10, 1 Layer
Tall Pillar Pitch	150	150
Bridge Die Thickness	55	55

Source: ASE

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2nd tier OSATs: most noticeable updates from other OSATs at SEMICON Taiwan is from Powertech on the panel level packaging where the company claims its 510x515mm panel level solution has improved from sub-panel throughput in '25 to full grid panel in '26, allowing 5.5x reticle package size application output improving from 8 units on a 300x300mm wafer to 36 units in its sub-panel offering further to 45 units in the full-grid solution.

The company aggressively targets its panel level packaging solutions to support 20x reticle size packaging applications by 2030 (vs. current 3-5x and 9x in 2027-28). We would note Powertech's FOPLP platform is able to support the logic die size up to 1x reticle size but the complexity of RDL stacking it can support is only up to 5 layers with limited bridge capability, still behind TSMC and ASE's technology portfolio to compete in the complex AI chipsets.

Supply/demand: On strong AI demand, the back-end equipment/materials suppliers see an acceleration of the 2.5D packaging capacity expansion led by TSMC and ASE followed by Amkor and Powertech, similar to [our: recent feedback](#), with All Ring contracting more business while AblePrint targeting to double its capacity again in '27.

3D packaging: heterogenous integration drives adoption inflection but requires closer supply chain collaboration

TSMC highlighted its 6um hybrid bond pitch in mass production since 2025 and targets narrowing it to 4.5um by '29, more than 50x interconnect density and ~5x energy efficiency vs. conventional interconnect. The company shared at SEMICON Taiwan



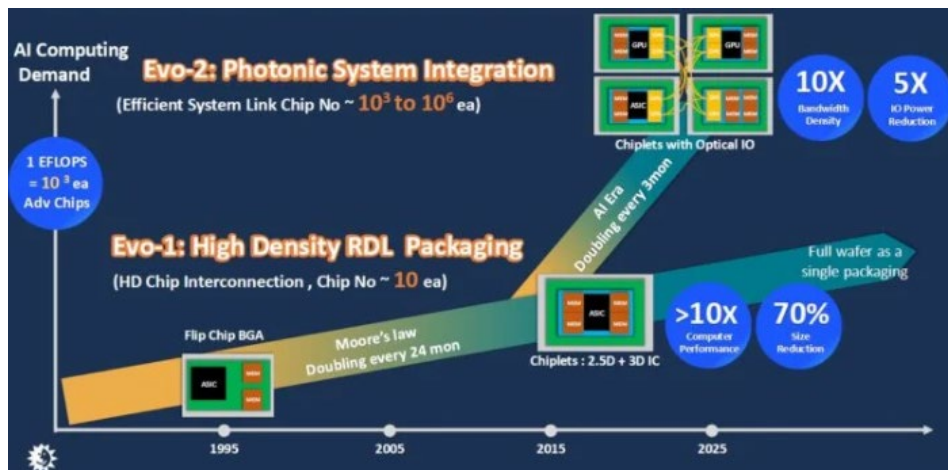
keynotes that N2P on N3P 3D integration will enter production in 2026 and it will have A14-on A14 stacking ready in 2029 while reiterated the integrated SoIC and CoWoS system will improve the compute capability by 50x in 2029 vs. level in 2024.

On the 3D packaging use case in CPO, TSMC noted the 2-4um bond pitch holds transmission loss at 112Gbps to just 0.05dB (vs. 1.38dB for a conventional micro-bump interconnect), bringing latency down to 10-20 nanoseconds. The company has a full optical PDK supporting Cadence and Synopsys' co-simulation and wafer level optical testing for edged coupled devices to screen defective dies before CPO assembly, improving the cost efficiency.

As more high value dies integrated vertically on top of the 2.5D base into a single 3D package to deliver shorter interconnect, higher bandwidth and lower energy per bit, TSMC pointed "1% yield loss is a shocking economic impact". The key technology gaps which could hold back chiplet and 3D packaging in mass production identified by the company would be 1) warpage from Si/organic CTE mismatch, 2) substrate fabrication limits, 3) large die handling stress / crack risk and 4) rising multi-material cost, tool automation gaps and 5) software integration.

Exhibit 24: ASE's advanced packaging roadmap

2.5D/3D packaging and optical I/O would be the trend to meet future AI computing chips requirements



Source: ASE

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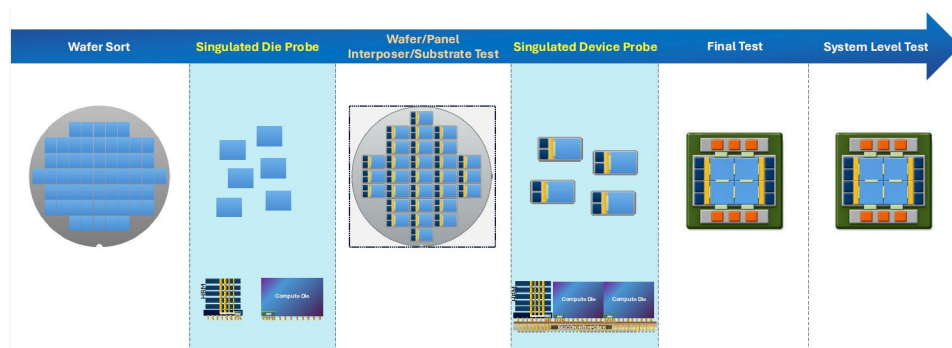
Testing / testing interface: growing importance in wafer and system level testing with benign supply/demand

The presenters at Advanced Testing Forum at SEMICON Taiwan including TSMC, KYEC, Advantest and Teradyne confirmed reliability is no longer a chipset/device level issue and testing process needs to move upstream and connect design, manufacturing and system level validation. This is especially important as training and inference accelerator chipsets and systems are getting more expensive with higher levels of design and manufacturing complexity, requiring extra testing to ensure consistency and durability.

Specifically, the supply chain sees the advanced packaging roadmaps in power, size and heterogenous integration are driving testing upgrades across the ATE, prober and handler equipment as well as the testing interface technologies. The testing coverage and importance are also being shifted to the left (wafer level) from right (device level) as the manufacturers try to ensure higher yield in interposer, substrate and wafer/panel level before assembly for better cost and efficiency control.

Exhibit 25: Test steps across chip manufacturing

Additional testing steps are proposed for enhanced control to address the reliability concerns especially for HPC/AI chipsets



Source: Teradyne

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To address the reliability concerns especially for HPC/AI chipsets, the additional testing steps are proposed for enhanced control including 1) singulated die probe (after wafer sort and before interposer/substrate test) to identify known good dies with enhanced stress which allow the supply chain to catch defects before they consume downstream materials and 2) singulated device probe (after interposer/substrate test and before final chipset level test) to verify die-to-die interconnect and incremental core coverage.

Interestingly, CPO wafer level testing was one of the key topics widely discussed at SEMICON Taiwan as the traditional optical test is too slow for mass deployment. In the CPO era, the testing service providers for insertion 2 will need to break 60+ years of single sided wafer testing convention to double sided electro-optical test. They will need to get light in & out, requiring a fiber to be aligned to an on-wafer coupler to sub-micron precision which takes minutes in lab. On top of the lengthy pre-testing process, the testing cycle itself is also long as all of the tests across optical signals, electrical signals and thermal reliability are all completed in one insertion.

For the AI applications, the semiconductor industry is seeing testing transforming from a cost center to a revenue enabler as test is only a small share of device/system cost which is far less than wasting constrained capacity elsewhere. On the other hand, every scrap not only carries a high cost and reduces revenue that can be shipped but also dampens the suppliers' commitment in delivery and quality control. As the testing value content grows along with testing technology migration and integration across hardware (e.g. equipment) and software (e.g. AI assisted analysis) while testing cycle takes longer with more stations required, we expect the semiconductor testing supply chain to benefit from the trend.

While the testing flow of CPO is largely defined by the key GPU customer from US, we reiterate our view that it does not make sense to have single source (supplier) in the individual testing stage, in the long run. Further, there could still be additions and/or deletions of testing steps in CPO going forward, along with the standardization of testing process.

For MPI, the qualification in Insertion 2 remains under way, while it is on track to ship the engineering sample of Insertion 3 prober in 4Q26. We also gained confidence from the company's total solution and strong after-sales services in the probe card business. As for WinWay, it has involved with more than 10 CPO-related on hand, and the pin count of testing socket can go up to 30+k based on the customers' roadmap. On HyperSocket, we note the key know-how is not on the rubber material, but instead it is more related to the formation of each piece of elastomer (on the socket to contact with the device).



Across the board, we see supply chain bottlenecks remain for testing interface names vs strong demand forecast by customers, and these include the labor force (for assembly in probe card and equipment) and ~12 months lead time of certain equipment (e.g. drilling for mechanical parts).

Back-end equipment: automation, process and metrology in focus

The major advanced back-end packaging suppliers at SEMICON Taiwan all noted factory space and equipment availability are two of the most severe manufacturing bottlenecks constraining their capability to meet the customers' demand at least through 2027, with ASE simultaneously building 7 buildings in Kaohsiung area alone. Based on the supply chain feedback at SEMICON Taiwan, the equipment delivery schedule is extending due to tight material supply and labor shortage, with wafer level process related tool lead time now beyond 9-12 months (vs. normal 2 quarters) while also getting longer for the chip level packaging process and testing equipment, suggesting strong business visibility for the back-end equipment supply chain into '27-28.

Longer term, the much shorter cadence of technology migration vs. the past few decades due to the accelerating AI upgrades is also putting pressure for the supply chain as the development has been shifting from serial to parallel development across chip, package, materials and system players. This requires the system integrators' joint-effort in the 3D IC Alliance (3DICAMA) as the component level validation along can no longer certify package once more components are tightly integrated.

TSMC at SEMICON Taiwan has emphasized the importance of higher level of back-end tool automation which is lag front-end fab equipment in software integration and maturity. Specifically, it noted new 2.5D packaging tools required to score at least 90 on automation based on the standard including the automation specs, test interfaces and emulator software it targets to release its partners by 4Q26. Separately, for the packaging process, TSMC and ASE will also screen the feasible solutions to address the challenges including warpage, large package size, process precision, cost, automation and labor in 2027 and convert them into actual adoption plans from 2028.

The other area the equipment supply chain will need to enhance their back-end to enable faster back-end packaging technology upgrades would be metrology and inspection where TSMC is unifying defect classification schemes and data formats across process steps so the data could be linked for AI image analysis consistently and it targets to bring AI based inspection and real time process control directly onto production lines in 2028.

To deliver better progress in advanced packaging technology innovation, TSMC announced its Baipu plan at SEMICON Taiwan. Out of the 53.6 hectares of industrial-use zone in Baipu Kaohsiung, 25 of those hectares are earmarked for TSMC and its supply chain. TSMC's own site runs to 3 hectares across two buildings on a modular layout so the space and resource can be reconfigured as technology evolves rather than requiring new construction each generation. Based on the current plan, the company will house small cleanrooms and labs functioning as a technology validation engine for the next generation advanced packaging technology, tools and materials.

Specifically, the plant will cover key CoWoS/CoPoS process stations with tool and material vendors encouraged to establish their own labs in the same building. By co-locating the vendors next to the validation line, it expects it to compress the prototype tool feedback loop from weeks/months to hours and a 25-50% improvement in validation efficiency, with total tool/material validation compressed to within 1 year.

On the company level, the equipment makers' feedback indicates the warpage remains the gating issue as more RDL layers stack with different materials. Alliance Material is addressing it with its balanced film through elastic modulus and CTE matching to absorb and distribute the thermal stress uniformly and has seen the solution gaining traction

from OSATs and foundries for meaningful ramp into '27-28. AblePrint with expertise in baking ovens used to collapse the micro-voids through high temperature and pressure now also has a warpage suppression system which is suitable for panel level packaging and tray processing.

CPO: production ramping but not maturing until 2H27; testing stages and share remain in debate

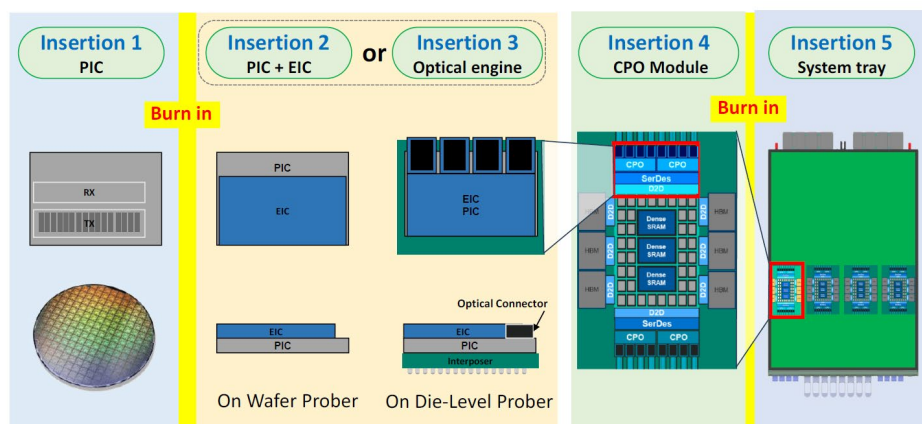
The industry is seeing interconnects as one of the key constraints on AI performance and thus the related upgrade is necessary to unlock the full potential of the logic chipsets and memory. Specifically, Lightmatter claims based on the same GUP under the same power environment, the adoption of photonic scale-up vs. traditional interconnect could shorten the training time by 3x while doubling the performance in decoding for inference tasks. As the industry is pushing toward ~1000 GPU scale up domain, the optical networking solutions could reduce the programming and communication constraints associated with partitioned systems, making the interconnected hardware behave more like a single machine.

From the mass production perspective, the experts from the leading companies acknowledge CPO is not yet fully ready in the supply chain at least through 1H27 as the test capacity and time along with the coupling throughput and precision are among the unresolved issues at this stage. However, during the transition, they see NPO would be a nearer term commercialization opportunity, still requiring the equipment readiness for more meaningful deployment and consistent with the supply chain feedback that the capacity ramp could happen 6-12 months ahead of the mass deployment.

All Ring at our booth tour noted it is extending its interposer/packaging equipment platform toward fiber attached and CPO assembly with its equipment throughput reaching 50-60 units per hour for current generation product. The solution will have a meaningful upgrade in 2027 to be even more competitive against ADST's solutions from both throughput and precision perspectives supplemented by its strength in algorithm, software and fully automated integrated system with solid feedback from its direct fabless customers and foundry and OSAT partners.

Exhibit 26: CPO insertion 1 to insertion 5 testing steps

Test capacity and time along with the coupling throughput and precision are among the unresolved issues



Source: ISE Labs

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From the testing equipment supply perspective, as the manufacturers get the capacity ready for higher level of photonic interconnect adoption, FiconTEC is targeting to ship ~1,000 systems in the next 12-18 months (vs. 2,000 tools in the past 25 years). The bottleneck for the industry is the shortages of expertise in photonic engineering, topical architecture, fiber coupling, precision alignment and test methodologies.

Specifically, when discussing on the insertion 2 double-sided electrical and optical testing after the wafers are integrated, the company noted it has automated probers already in shipment which are already compatible with Teradyne's testers while the configuration to support Advantest's testers will be ready for shipment in October this year, putting pressure on the late comers including MPI whose solutions are still under qualification.

Exhibit 27: CPO insertion 1-3 tool suppliers

The bottleneck for the industry is the shortages of expertise in photonic engineering, topical architecture, fiber coupling, precision alignment and test methodologies

Testing process	Tool suppliers and models	Broad deployment	ATE interface	No cables interface	High throughput	Test count support
Insertion 1	Formfactor CM300	Proven	Not demonstrated	Not demonstrated	Not demonstrated	Not demonstrated
	TEL/Advantest/Formfactor Triton	Emerging but limited	Proven	Not demonstrated	Emerging but limited	Not demonstrated
	Technoprobe Eclipse	Not demonstrated	Proven	Proven	Emerging but limited	Proven
Insertion 2	FiconTec WLT-02	Not demonstrated	Proven	Proven	Emerging but limited	Emerging but limited
	MPI TS3500-DS	Not demonstrated	Proven	Proven	Emerging but limited	Emerging but limited
Insertion 3	FiconTec DLT-D1 INS 3	Not demonstrated	Not demonstrated	Not demonstrated	Proven (3 sites)	Proven
	MPI DTS650-DI	Not demonstrated	Proven	Proven	Emergin (Multi-sites in 2027)	Proven
	Chroma	Not demonstrated	Proven	Proven	Proven	Proven

Source: BofA Global Research

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On the insertion 3 die level electro-optical testing for optical engines, FiconTEC also noted it has been shipping the equipment to its OSAT partners for more than six months since 1Q26, with 250k OEs already being tested. However, the supply chain feedback from Chroma indicated that it has been gaining share in insertion 3 testing for the optical engines at the expense of FiconTEC when the supply chain moved from R&D stage to pilot run. It believes its share could stay multiple times larger than its competition when the supply chain starts pulling in equipment for mass production based on the positive feedback on testing quality and cycle from foundries and OSATs.

On the later testing processes, while the testing specs for NPO/CPO are not yet fully finalized, MPI and Winway are targeting optical/electrical probing and sockets ready for photonics ICs and CPO modules by the end of this year. Hon Precision has been providing the electric testing handlers for Nvidia's Spectrum and Quantum solutions and targets to start shipping the engineering equipment in October for the networking switch ASIC electrical testing and bonded-OE optical testing though the qualification and mass production schedule still depends on the customers specs finalization.

Mediatek: Nvidia/Alphabet partnership reassures ASIC opportunity

Mediatek completed US\$3.9bn ECB issuance, with conversion price set at NT\$4,513.75 (15% premium to Aug 31st closing) and US\$3.5bn subscribed by Nvidia, followed by Alphabet and institutional investors. The shareholder dilution is limited at 1.67% if the ECB is fully converted by 2031. [We estimate](#) the ECB proceeds will boost its net cash to ~US\$14bn, allowing it to lock in the components where the supply is tight through long term agreements (e.g. HBM and substrate) for its ASIC customers requiring the service.

Beyond the collaboration in PC/workstation and automotive, Nvidia's direct investment demonstrates its confidence in Mediatek's capability in AI infrastructure integrating the hyperscalers/enterprises' custom XPUs to Nvidia's rack-scale system through NVLink Fusion ecosystem in chiplets, chip-to-chip interconnect and custom HBM. Along with pre-validated hardware, the partnership allows seamless connection between chipsets and components at lower system design complexity and faster time to market for semi-custom AI factories. While our base case for Mediatek's ASIC sales will ramp from US\$2bn in '26 to US\$48bn in '28 led by TPUs supplemented by xAI's accelerator chipset, the company should have flexibility to expand its addressable market to components and hardware which carries 5-10x ASPs vs standalone chipsets as it provides the value add in designing, sourcing and system integration for 2nd tier hyperscalers and enterprises.

Specifically on TPU opportunity, similar to its recent comment at our Tech Forum in mid August, Mediatek shared that it expects the chipset it designs for Google TPUv8t will have relatively higher unit share when it starts ramping production from 4Q26 through the project lifecycle vs. Broadcom's TPUv8i. The company also expects the tape out of TPUv9 is well on track for tape out by the end of this year as multiple test vehicle milestones have been achieved, putting the chipset production well on track by late '27 / early '28 with the majority of the share despite the concerns on the design progress and Intel's EMIB-T yield learning curve.

For TPUv10, while the part of the computing content could be captured by different design service vendors and Google if the hyperscaler decides to tape out the die itself, Mediatek believes it should still own the I/O and base die design along with chiplet integration, allowing it to still grow the chipset dollar content which continues to grow its reticle size to 14x in 2029 (vs. 9-10x for TPUv9).

Exhibit 28: Terms and definitions

Glossary of abbreviations/acronyms

Term	Definition
AI	Artificial Intelligence
ASIC	Application Specific Integrated Circuit
ATE	Automated Test Equipment
BOM	Bill of Materials
CMP	Chemical-Mechanical Polishing
CoPoS	Chip-on-Panel-on-Substrate
CoWoS-L	Chip-on-Wafer-on-Substrate-Local Silicon Bridge
CoWoS-R	Chip-on-Wafer-on-Substrate-Redistribution Layer
CPO	Co-Packaged Optics
CPU	Central Processing Unit
CTE	Coefficient of Thermal Expansion
DRAM	Dynamic Random Access Memory
ECB	Euro-Convertible Bond
EDA	Electronic Design Automation
FoCoS	Fan-Out Chip on Substrate
FOPLP	Fan-Out Panel Level Packaging
GPU	Graphic Processing Unit
GWC	GlobalWafers
HBM	High Bandwidth Memory
HPC	High-Performance Computing
I/O	Input/Output
IC	Integrated Circuit
IP	Intellectual Property
KYEC	King Yuan
NPO	Near-Packaged Optics
OSAT	Outsourced Semiconductor Assembly/Testing
PDK	Process Design Kit
RDL	Redistribution Layer
SiC	Silicon Carbide
SoIC	System on Integrated Chips
SWIFT	Silicon Wafer Integrated Fan-out Technology
TPU	Tensor Processing Unit
WPM	Wafer per month

Source: BofA Global Research

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Exhibit 29: Stocks mentioned

Prices and ratings for stocks mentioned in this report

BofA Ticker	Bloomberg ticker	Company name	Price	Rating
XNQNF	6187 TT	All Ring	NT\$ 1370	C-1-7
XSRIF	3711 TT	ASE Technology	NT\$ 621	C-1-7
CRMJF	2360 TT	Chroma ATE	NT\$ 2230	C-1-7
GBUHF	3443 TT	Global Unichip Corp.	NT\$ 5935	C-1-7
MDTKF	2454 TT	MediaTek	NT\$ 4760	C-1-7
TSMWF	2330 TT	Taiwan Semi Mfg Co	NT\$ 2460	B-1-7



Exhibit 29: Stocks mentioned

Prices and ratings for stocks mentioned in this report

BofA Ticker	Bloomberg ticker	Company name	Price	Rating
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Source: BofA Global Research

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Price objective basis & risk**All Ring (XNQNF)**

Our PO of NT\$1,750 is based on 36x 2027E P/E, positioned at the upper half of its historical trading range of 8-41x. We believe P/E based valuation is appropriate given All Ring's robust top-line visibility, stable profitability profile, and structural earnings improvement on decent exposure to AI-driven advanced packaging upcycle. The valuation multiple is underpinned by

1) 34%/37% CAGR in sales and earnings over 2026-28E and 2) sustained ROE in the 30-40% range

Downside risks are:

1) slowdown in advanced packaging capacity expansion and 2) mis-execution on equipment development

ASE Technology Holding (XSRIF)

We value ASE Technology Holding at NT\$800 per share (US\$51 for ADR), based on 20x 2H27-1H28E P/E, above its historical range of 5-26x, in view of its solidified industry position in AI era. We use P/E ratio to value ASE as the company has been able to maintain healthy profitability through the cycles in the past 20 years due to consolidating industry landscape supporting a more benign supply/demand and keeping pricing and margins less cyclical.

Downside risks to our PO are 1) share loss and/or ASP erosion owing to Chinese players' cannibalization and/or ASE's worsening execution, 2) weakening end demand due to uncontrollable matters, such as macro halt and/or geopolitical tensions, 3) regulatory issues that could hinder ASE from generating synergies and benefit its competitors.

Chroma ATE (CRMJF)

Our price objective of NT\$3,050 on Chroma is based on 38x 2027E P/E, which is at the top of the company's historical trading range of 12-39x. In our view, the valuation multiple is underpinned by

1) +70% earnings CAGR in 2025-27E supported by +75% sales CAGR growth and GMs/OpMs expanding, 2) ROE expansion to around 35% in 2025-27E, 3) net cash of around NT\$21/share, and 4) free cash flow generation of around NT\$15/share per annum across 2025-27E.

Downside risks to our price objective are:

1) weaker market demand, especially the cloud AI investment moderation, leading to slower infrastructure builds especially in China and dragging the company's revenue growth for its power testing equipment business and operating leverage
2) a more conservative investment cycle from the hyperscalers on further data center deployment due to growing capex could dampen the demand for GPU and ASIC, negatively impacting the testing equipment demand for related chipsets.



3) failure to execute on the customized equipment requested by its customers or delay of technology roadmap could lead to an order shift to its competitors.

Global Unichip Corp. (GBUHF)

Our PO NT\$7,000 is based on 50x 2H27-1H28E P/E. This PE multiple is around the high-end of the historical trading range of 10-70x from 2013-2026, on the back of: 1) around 88% earnings CAGR during 2026-28E, 2) 46% ROE during 2026-27E, 3) robust cash levels, 4) operating margin of 15% into 2026-28E.

Upside risks are

- (1) stronger than expected revenue from substrate design business
- (2) stronger for longer demand from existing mass production projects, e.g. SSD (solid state drive) controller, networking chip

Downside risks:

- (1) further delay of NRE (non recurring engineering) projects which could subsequently lead to the delay in entering into turnkey stage (and thus slower operating margin expansion)
- (2) sudden shutdown or halt in China-related business given the geopolitical tensions
- (3) effects of low-margin cryptocurrency ASIC wind down are worse than expected
- (4) Weaker than expected shipment of Google's TPU, leading to lower turnkey revenue contribution
- (5) Unexpected delay in ADAS project development, leading to slower ramp in revenue contribution from 2027

MediaTek (MDTKF)

Our PO of NT\$5,230 is based on 23x 2H27-1H28 P/E. as we are constructive on its broadening ASIC pipeline to drive earnings upgrade cycle and re-rating as it locks in more supply.

Upside risks to our PO are: (1) better than expected smartphone demand, especially in China, (2) a better pricing in mobile communication segment thanks to alleviated competition and rising exposure to high-end/flagship market, and (3) an even more superior investment return on non-op side to uplift its net cash position. (4) Unexpected collaboration with global technology leaders leads to higher than expected margins/profits

Downside risks to our PO are: (1) weaker-than-expected end-market situation in consumer-related smartphone/TV space, (2) macro challenges including inflation/policy-related pressures, (3) rising competition from existing players and/or newcomers in smartphone/TV SoC (system on chip) space, and (4) ASP/margin pressure caused by each or all of the above and/or rising foundry costs. (4) Bidding process for Meta Inference ASIC is slower than expected.

Taiwan Semiconductor Manufacturing Co. (TSMWF)

Our PO for TSMC is NT\$3,100 per share (US\$590 per ADR), based on 20x 2027E P/E. TSMC's valuation has traded in the range of 10-26x PE from 2017. 20x is in the mid to high end of the historical range, above the average P/E of 18x. We think a PE above the average is justified given TSMC's enhanced industry position and mid-40% 5-year CAGR in AI should underpin a rerating

Downside risks to our price objective are: (1) greater-than-expected slowdown in global smartphone/consumer electronics demand, (2) Intel's potential insourcing strategy and ambitions in foundry service, (3) execution risks on advanced nodes, and (4) higher than expected tariffs.



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BUY				
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	All Ring	XNQNF	6187 TT	Haas Liu
	AMEC	XRTOF	688012 CH	Dai Shen
	ASE Technology Holding	XSRIF	3711 TT	Haas Liu
	ASE Technology Holding -ADR	ASX	ASX US	Haas Liu
	ASMPT	ASMFV	522 HK	Simon Woo, CFA
	Aspeed	XLKMF	5274 TT	Mike Yang
	Chroma ATE	CRMJF	2360 TT	Haas Liu
	Doosan Corp	DOSAF	000150 KS	Simon Woo, CFA
	eMemory	XYLWF	3529 TT	Mike Yang
	GigaDevice	XGXIF	603986 CH	Daley Li, CFA
	Global Unichip Corp.	GBUHF	3443 TT	Haas Liu
	Grand Process Technology Corp	XZOWF	3131 TT	Haas Liu
	Hanmi Semiconductor	HNSIF	042700 KS	Simon Woo, CFA
	Hon Precision	XXPIF	7769 TT	Haas Liu
	Horizon Robotics	HRZRF	9660 HK	Daley Li, CFA
	Iluvatar CoreX	XSIBF	9903 HK	Daley Li, CFA
	Ingenic	XISCF	300223 CH	Dai Shen
	InnoScience Technology	XSCHF	2577 HK	Daley Li, CFA
	JCET Group Co Ltd	XJIEF	600584 CH	Dai Shen
	King Yuan Electronics Corp.	KYUFF	2449 TT	Haas Liu
	Kinik	KIKCF	1560 TT	Haas Liu
	Macronix International	MXICF	2337 TT	Dai Shen
	MediaTek	MDTKF	2454 TT	Haas Liu
	Montage Technology	XRDF	688008 CH	Daley Li, CFA
	MPI Corporation	XMJCF	6223 TT	Mike Yang
	Nanya Technology	NNYAF	2408 TT	Simon Woo, CFA
	NCE Power	XVFFF	605111 CH	Daley Li, CFA
	OmniVision	XXHQF	603501 CH	Dai Shen
	Phison Electronics	PISNF	8299 TT	Simon Woo, CFA
	Powerchip Semiconductor Manufacturing Co	XCHPF	6770 TT	Haas Liu
	Powertech Technology	PWOGF	6239 TT	Simon Woo, CFA
	Rockchip	XRPXF	603893 CH	Daley Li, CFA
	Samsung C&T	XCDTF	028260 KS	Simon Woo, CFA
	Samsung Elec -G	SSNHZ	SMSN LI	Simon Woo, CFA
	Samsung Electronics	SSNLF	005930 KS	Simon Woo, CFA
	Samsung Electronics Preferred	SSNNF	005935 KS	Simon Woo, CFA
	Silergy Corp.	SLEGF	6415 TT	Mike Yang
	Silicon Motion	SIMO	SIMO US	Simon Woo, CFA
	SK Hynix	HXSCF	000660 KS	Simon Woo, CFA
	SK Hynix ADR	SKHY	SKHY US	Simon Woo, CFA
	SK Square	SKSQF	402340 KS	Simon Woo, CFA
	Taiwan Semiconductor Manufacturing Co.	TSM	TSM US	Haas Liu
	Taiwan Semiconductor Manufacturing Co.	TSMWF	2330 TT	Haas Liu
	Vanguard International Semiconductor Co	VGILF	5347 TT	Haas Liu
	Winbond Electronics	WBEKF	2344 TT	Dai Shen
	WinWay Technology	XWCLF	6515 TT	Mike Yang
	WT Microelectronics	XZOPF	3036 TT	Mike Yang
NEUTRAL				
	ASMedia Technology Inc.	XZSFF	5269 TT	Mike Yang
	Black Sesame Intl Holding	BSIHF	2533 HK	Daley Li, CFA
	Hangzhou Silan Microelectronics	XDFRF	600460 CH	Daley Li, CFA
	Lion Electronics	XDHFF	605358 CH	Dai Shen
	MetaX	XMEZF	688802 CH	Daley Li, CFA
	Novatek	NVKMF	3034 TT	Mike Yang
	Realtek	RLTKF	2379 TT	Mike Yang
	Shenzhen Goodix	XQPLF	603160 CH	Daley Li, CFA
	Soulbrain	XSBOF	357780 KS	Simon Woo, CFA
	VeriSilicon	XMLZF	688521 CH	Daley Li, CFA
UNDERPERFORM				
	Crystal Clear	XPPTF	300655 CH	Dai Shen
	Faraday	FDYTF	3035 TT	Haas Liu
	GlobalWafers	XWLFF	6488 TT	Mike Yang



APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	Hua Hong Grace	HHUSF	1347 HK	Dai Shen
	LX Semicon	XLXS	108320 KS	Simon Woo, CFA
	M31 Technology	XMTZF	6643 TT	Mike Yang
	Maxscend	XMXSF	300782 CH	Daley Li, CFA
	Parade	PRDWF	4966 TT	Mike Yang
	United Microelectronics Corp.	XUMIF	2303 TT	Haas Liu
	United Microelectronics Corp.	UMC	UMC US	Haas Liu
	Win Semiconductors	WSMIF	3105 TT	Haas Liu
	Wonik IPS	XRHQF	240810 KS	Simon Woo, CFA

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Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	243	60.60%	Buy	123	50.62%
Hold	87	21.70%	Hold	45	51.72%
Sell	71	17.71%	Sell	21	29.58%

Equity Investment Rating Distribution: Global Group (as of 30 Jun 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	1987	56.11%	Buy	1190	59.89%
Hold	797	22.51%	Hold	496	62.23%
Sell	757	21.38%	Sell	391	51.65%

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Buy	≥ 10%	≤ 70%
Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

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